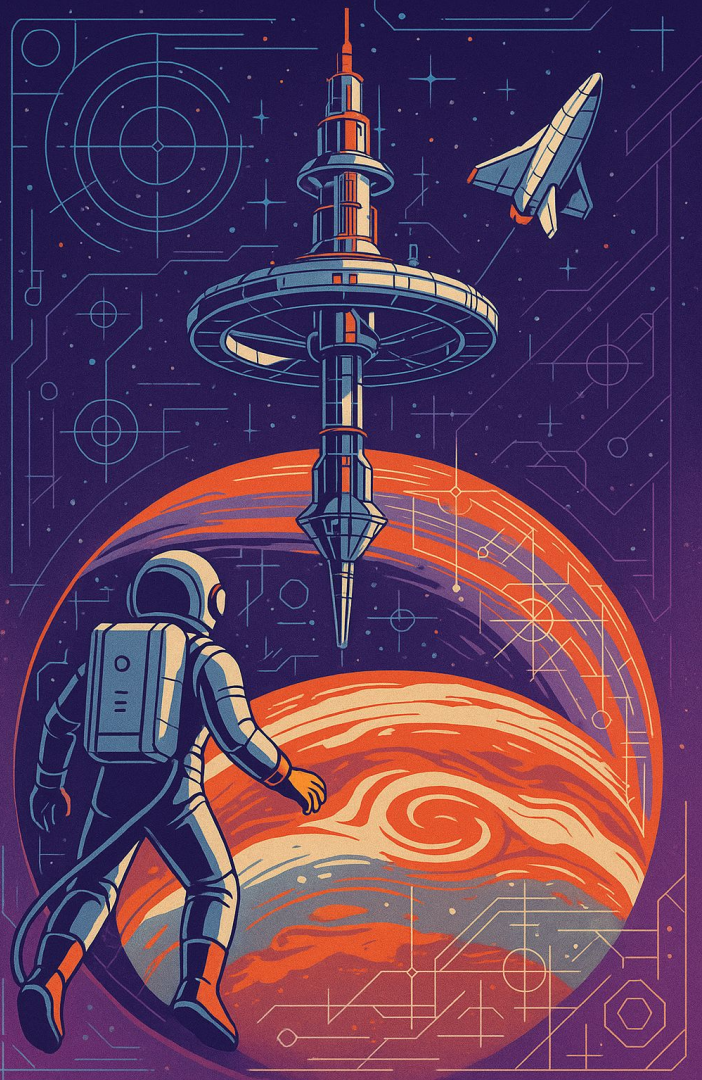




Gen AI and Agentic AI in Production

Week 4
Day 1

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

GitHub
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

Agent Architectures

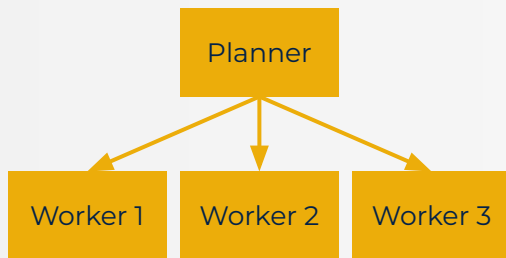
No standard terminology, but Anthropic's definitions:

Workflows are systems where LLMs and tools are orchestrated through predefined code paths.

Agents are systems where LLMs dynamically direct their own processes and tool usage, maintaining control over how they accomplish tasks.

2 common patterns for Agent architectures:

Multi-Agent



Single Agent with Loop



Alex - the financial planner



What it is:

SaaS commercial application, financial planner

Done:

Researcher Agent with Data Ingest

Todo:

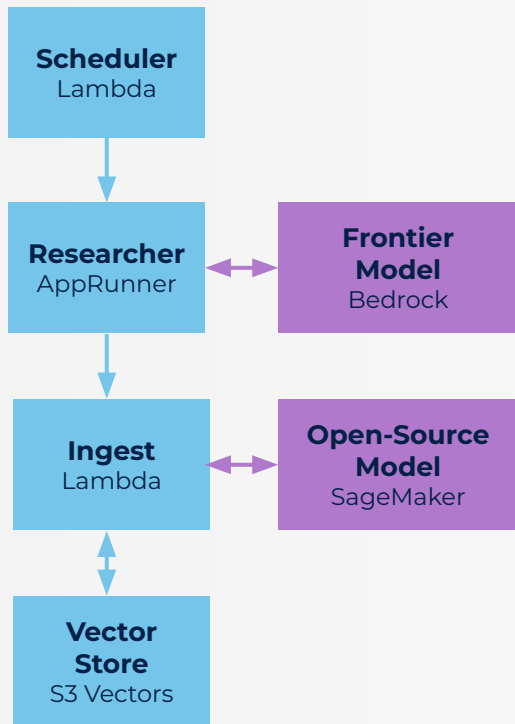
Day 1: Database

Day 2: Five Agents

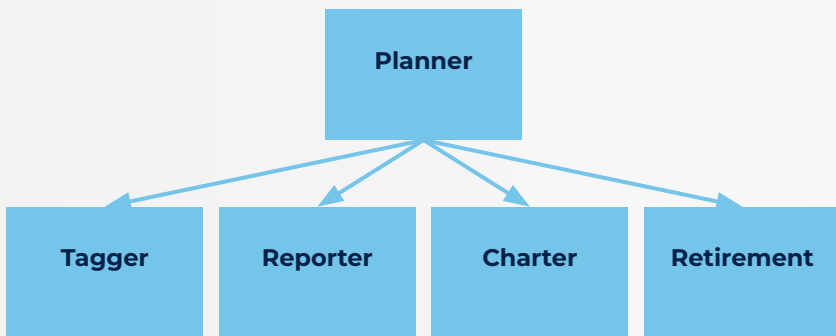
Day 3: Frontend + API

Day 4: Enterprise Grade

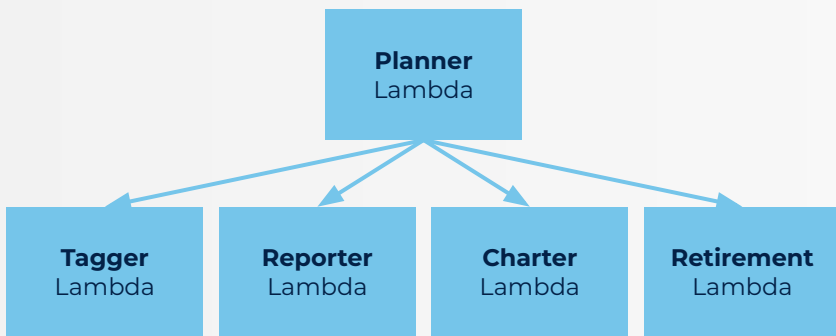
Data Ingest Architecture



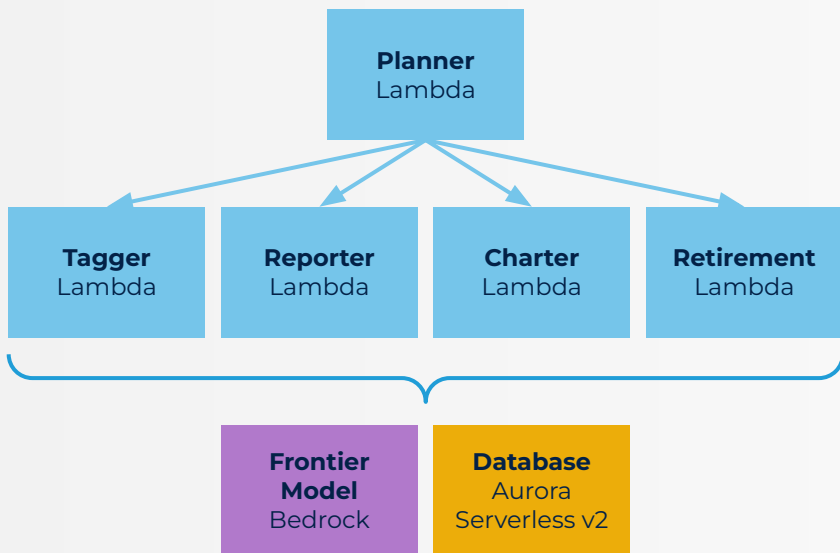
Multi-Agent Architecture



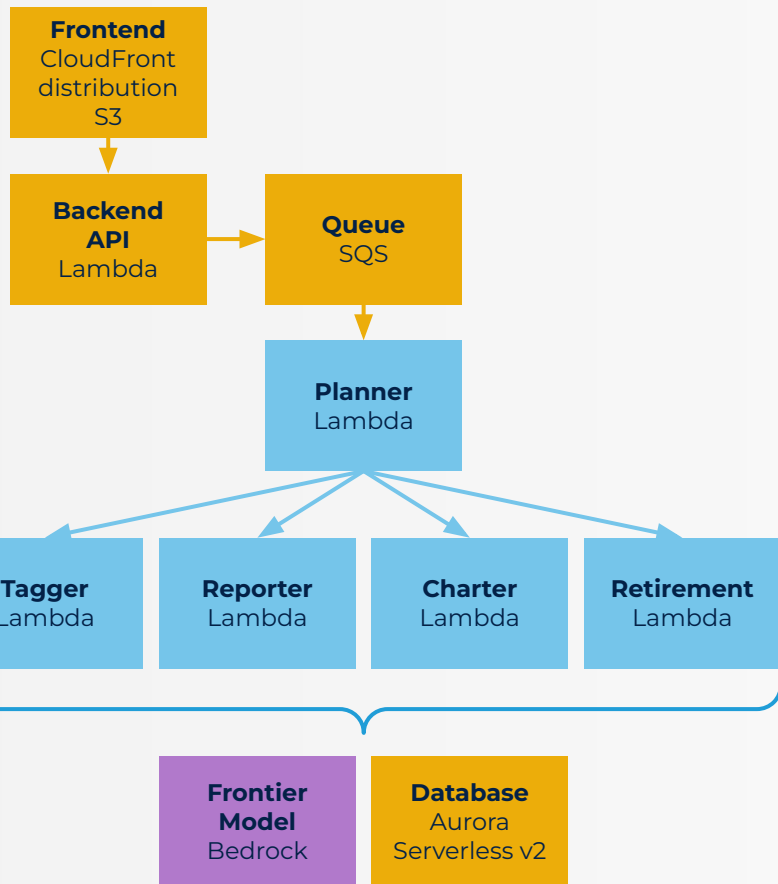
Multi-Agent Architecture



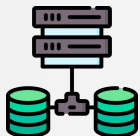
Multi-Agent Architecture



Multi-Agent Architecture



Relational Databases in AWS



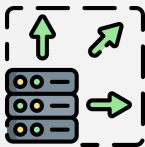
RDS

Relational Database Service: the umbrella managed service for relational databases, supporting different engines (MySQL, PostgreSQL, ...)



Aurora

One of the Database Engines offered in RDS; use RDS to create the database cluster, select Aurora as the engine. Designed by AWS to be fast and scalable.



Aurora Serverless V2

Automatically adjusts database capacity responding quickly to changes in workload without downtime for pay-as-you-go scaling

And a zoo of other non-relational db offerings beyond RDS -
eg, Dynamo is the NoSQL offering; ElastiCache; DocumentDB

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Gen AI and Agentic AI in Production

80%
Complete